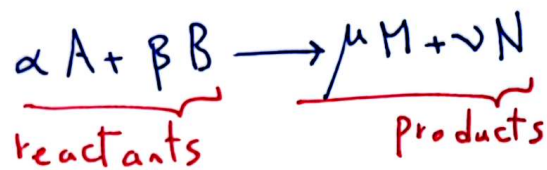


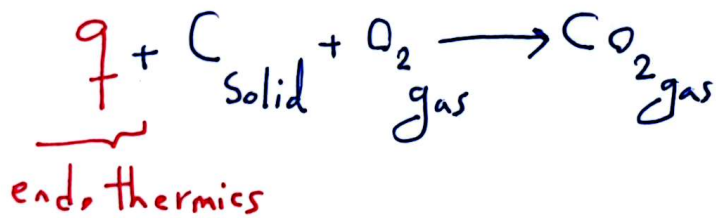
1 Combustion:

Combustion is usually considered to take place in chemical equilibrium.

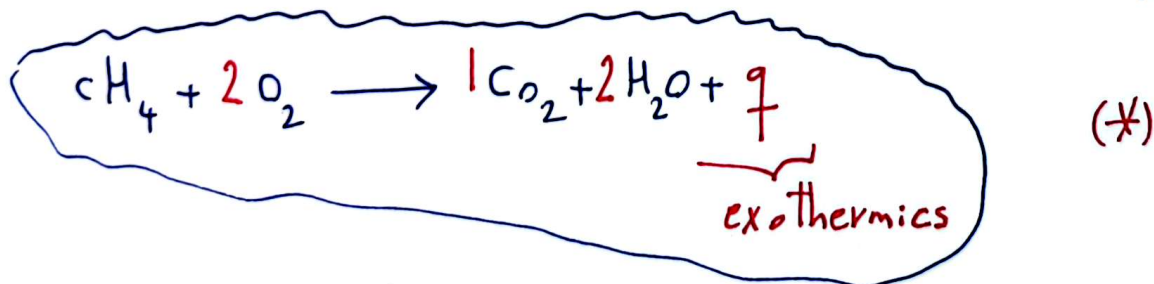
chemical equilibrium means that there is no-change in concentration of reactants & products.



If a chemical reaction absorbs heat to occur, it is called "endothermics". Ex)

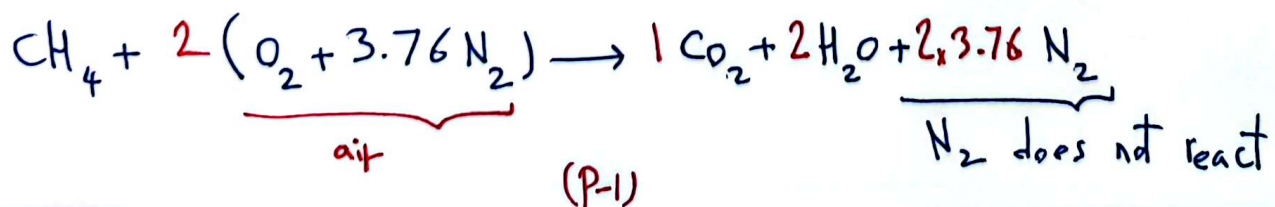


If the reaction releases heat, it is "exothermics".



Hydro-carbon fuels: $C_n H_{2n+2}$ n : number of carbons

Consider (*), the more realistic reaction is



Note: One mole of air has 3.76 times more N_2 than O_2 :

$$\frac{0.21 O_2 + 0.79 N_2}{0.21} \longrightarrow = (O_2 + 3.76 N_2)$$

Stoichiometric Ratio: It refers to exact air-to-fuel ratio needed for complete combustion (No excess air, no excess fuel in products)

$$AF) = \frac{\text{mass of air}}{\text{mass of fuel}}$$

Air-to-fuel
ratio

For the CH_4 reaction, $AF)_{sto} = \frac{2[32 + 3.76 \times 28]}{12 + 4} = \boxed{17.16}$

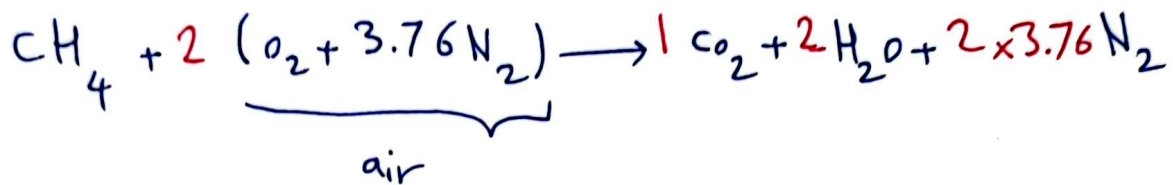
If $AF > AF_{sto} \longrightarrow$ reaction is air-rich (excess air)

If $AF < AF_{sto} \longrightarrow$ reaction is fuel-rich (excess fuel)

Define $r = \frac{\text{mass of } O_2}{\text{mass of fuel}}$

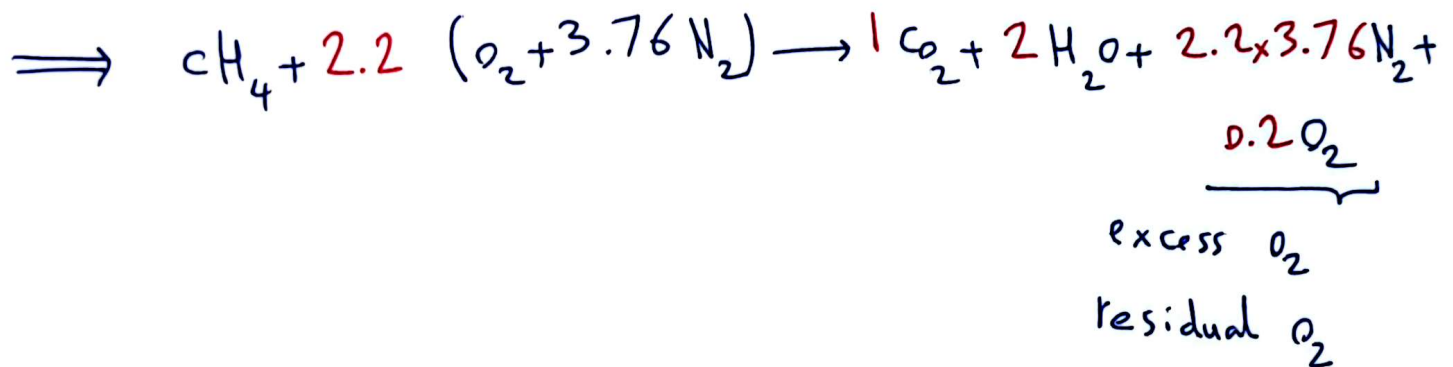
Example: We have 10% air-rich methane-air reaction, find coefficients of chemical reaction & write reaction.

Solution: start with stoichiometric reaction:



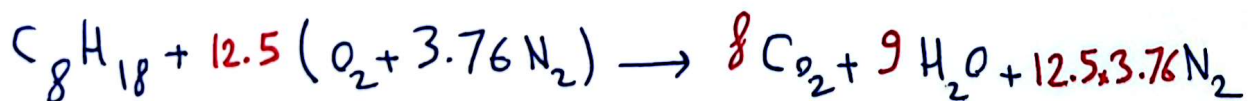
10% more $\rightarrow$ means 110% air or 1.1 more air

$$1.1 \times 2 = 2.2 \text{ (moles of air)}$$



Example: Gasoline reacts with air, compute stoichiometric ratio.

Solution: Gasoline is octane (C_8H_{18})

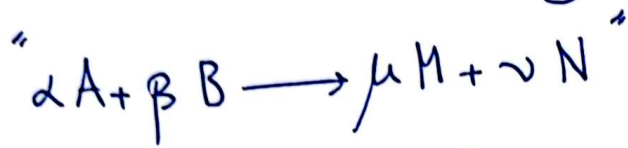


Balancing order: C - H - O - N

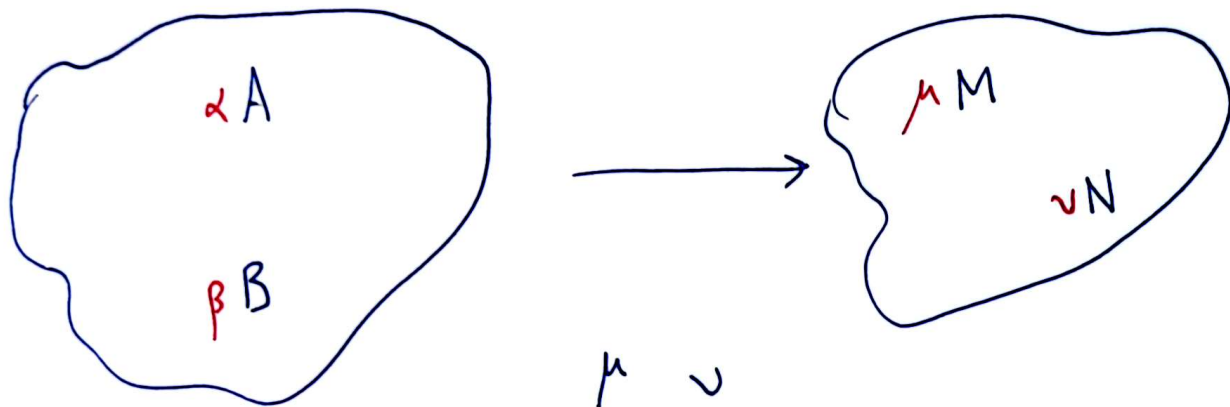
$$\text{AF})_{\text{sto}} = \frac{12.5(32 + 3.76 \times 28)}{8 \times 12 + 18 \times 1} = 15.25$$

"Equilibrium Constant"

Assume we have the following reaction:



It is shown that the following ratio is constant for the above chemical reaction:



$$K_p(T) = \frac{P_M^\mu P_N^\nu}{P_A^\alpha P_B^\beta}$$

Equilibrium constant
in terms of partial pressure

We can write Eq. constant in terms of mole fraction:

$$K_n(T) = \frac{X_M^\mu X_N^\nu}{X_A^\alpha X_B^\beta} \quad X: \text{mole fraction}$$

Relation between $K_n(T)$ & $K_p(T)$:

$$K_n = \frac{P_m^{\alpha+\beta-\mu-\nu}}{K_p}$$

mixture pressure

Adiabatic flame temperature:

When a reaction occurs in adiabatic condition, $\bar{H}_p = \bar{H}_R$, i.e., from first law, "enthalpy of products = enthalpy of reactants." If we solve this equation ($H_p = H_R$), we get temperature of products, which is called adiabatic flame temperature.

Example: Q-13 from chapter 2 of the textbook.

A mixture of air and $\text{CH}_{1.9}$ undergoes complete combustion. Twice as much air is present as needed to burn the fuel. How high is final temperature? ($T_R = 25^\circ$, use the below table)

Species	$\bar{Q}_f(\bar{h}_f) \frac{\text{kJ}}{\text{kmol K}} @ 298 \text{ K}$	$\bar{c}_p \left(\frac{\text{kJ}}{\text{kmol K}} \right)$
$\text{CH}_{1.9}$	8358	
$\text{CO}_2(\text{g})$	-393522	51.9
$\text{O}_2(\text{g})$	0	34.0
$\text{N}_2(\text{g})$	0	31.6
$\text{H}_2\text{O}(\text{g})$	-241827	40.6

$\bar{Q}_f(\bar{h}_f)$
Enthalpy of
formation at
298 K